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# Jiwon Lee

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Website | LinkedIn | Google Scholar | ORCID

I completed my Ph.D. at Yonsei University under the supervision of Professor Won Woo Ro, with a dissertation titled 'High-Performance Address Translation Mechanisms for Graphics Processing Units.' My research interests span a broad range of topics within computer architecture and systems, including (but not limited to) designing efficient CPU and GPU microarchitectures, virtual memory frameworks, scalable memory management methods. To broaden my experience, I am currently working as an engineer at Samsung Electronics.

## PROFESSIONAL EXPERIENCE

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| Staff Engineer              | Samsung Electronic Memory Division, Korea                                                                 | Aug. 2024 - Present   |
| Graduate Research Assistant | Yonsei University, Korea<br>Embedded Systems and Computer Architecture Lab (eSCaL)<br>Advisor: Won Woo Ro | Mar. 2014 - Aug. 2018 |

## EDUCATION

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| Doctor of Philosophy | School of Electrical and Electronic Engineering, Yonsei University<br>Dissertation: High-Performance Address Translation Mechanisms for Graphics Processing Units (Advisor: Won Woo Ro) | Mar. 2018 - Aug. 2024 |
| Bachelor's Degree    | School of Electrical and Electronic Engineering, Yonsei University                                                                                                                      | Mar. 2014 - Feb. 2018 |

## PUBLICATIONS

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### **LATPC: Accelerating GPU Address Translation Using Locality-Aware TLB Prefetching and MSHR Compression**

Yeonan Ha, Jiho Park, Hanna Cha, **Jiwon Lee**, Joonsung Kim, Won Woo Ro, and Youngsok Kim  
IEEE/ACM International Symposium on Microarchitecture (MICRO), 2025. (to appear)

### **Heliostat: Harnessing Ray Tracing Accelerators for Page Table Walks**

Yuan Feng, Yuke Li, **Jiwon Lee**, Won Woo Ro, and Hyeran Jeon  
IEEE/ACM International Symposium on Computer Architecture (ISCA), 2025.

### **COSMOS: An LLC Contention Slowdown Model for Heterogeneous Multi-core Systems**

Yongju Lee, Jaewon Kwon, Cheolhwan Kim, Enhyuk Jang, **Jiwon Lee**, Hyunwuk Lee, and Won Woo Ro  
IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS), 2025.

### **HashScape: Leveraging Virtual Address Dynamics for Efficient Hashed Page Tables**

Won Hur, **Jiwon Lee**, Jaewon Kwon, Minjae Kim, and Won Woo Ro  
IEEE Transactions on Computers (TC), 2025.

### **Marching Page Walks: Batching and Concurrent Page Table Walks for Enhancing GPU Throughput**

**Jiwon Lee\***, Gun Ko, Myung Kuk Yoon, Ipoom Jeong, Yunho Oh, and Won Woo Ro  
IEEE International Symposium on High-Performance Computer Architecture (HPCA), 2025.

### **REC: Enhancing Fine-Grained Cache Coherence Protocol in Multi-GPU Systems**

Gun Ko, **Jiwon Lee**, Hongju Kal, Hyunwuk Lee, and Won Woo Ro  
Journal of Systems Architecture (JSA), Vol. 160, 103339, Mar. 2025.

### **Geneva: A Dynamic Confluence of Speculative Execution and In-Order Commitment Windows**

Yanghee Lee, **Jiwon Lee**, Jaewon Kwon, Yongju Lee, and Won Woo Ro  
IEEE/ACM Design Automation Conference (DAC), 2024.

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### **SnakeByte: A TLB Design with Adaptive and Recursive Page Merging in GPUs**

**Jiwon Lee\***, Ju Min Lee, Yunho Oh, William J. Song, and Won Woo Ro

IEEE International Symposium on High-Performance Computer Architecture (HPCA), 2023.

### **Early-Adaptor: An Adaptive Framework for Proactive UVM Memory Management**

Seokjin Go, Hyunwuk Lee, Junsung Kim, **Jiwon Lee**, Myung Kuk Yoon, and Won Woo Ro

IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS), 2023.

### **Analysis on Memory Access Patterns of Server-Class Workloads in Page- and Cache Line- Granularity**

Kyeonghoon Lim, Minjae Kim, **Jiwon Lee**, and Won Woo Ro

International Conference on Electronics, Information, and Communication (ICEIC), 2023.

### **Performance Analysis of Criticality-Aware Out-of-Order Cores for Exploiting MLP**

Yanghee Lee, **Jiwon Lee**, and Won Woo Ro

International Technical Conference on Circuits/Systems, Computers and Communications (ITC-CSCC), 2023.

### **Adaptive Data Prefetcher with Probability Learning in LLC**

Jusin Kim, **Jiwon Lee**, and Won Woo Ro

International Technical Conference on Circuits/Systems, Computers and Communications (ITC-CSCC), 2023.

### **Reconstructing Out-of-Order Issue Queue**

Ipoom Jeong, **Jiwon Lee**, Myung Kuk Yoon, and Won Woo Ro

IEEE/ACM International Symposium on Microarchitecture (MICRO), 2022.

### **Analysis of DRAM-based Network of DRAM Swap Space Adopting Latency Hiding Technique**

Hyoseong Choi, **Jiwon Lee**, Jeonghoon Choi, and Won Woo Ro

International Technical Conference on Circuits/Systems, Computers and Communications (ITC-CSCC), 2022.

### **PIMCaffe: Functional Evaluation of a Machine Learning Framework for In-Memory Neural Processing Unit**

Won Jeon, **Jiwon Lee**, Dongseok Kang, Hongju Kal, and Won Woo Ro

IEEE Access, Vol. 9, pp. 96629-96640, Jul. 2021.

### **Chapter Six - Deep learning with GPUs**

Won Jeon, Gun Ko, **Jiwon Lee**, Hyunwuk Lee, Dongho Ha, and Won Woo Ro

Advances in Computers, Vol. 122, pp. 167-215, 2021.

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## **PATENTS**

### **Computing device and method of operating the same**

Won Woo Ro, Minjae Kim, Kyeonghoon Lim, and **Jiwon Lee**

KR 10-2023-0193178

### **Memory management apparatus and method for UVM**

Won Woo Ro, Seokjin Go, Junsung Kim, Hyunwuk Lee, and **Jiwon Lee**

KR 10-2023-0076606

### **Memory management unit and method of walking page table**

Won Woo Ro, **Jiwon Lee**, Ipoom Jeong, Hongju Kal, Gun Ko, and Hyunwuk Lee

KR 10-2022-0175909, US 18/502,058, CN 202311700848.0

### **Swap memory device providing data and data block, method of operating the same, and method of operating electronic device including the same**

Won Woo Ro, Hyoseong Choi, **Jiwon Lee**, and Jeonghoon Choi

KR 10-2022-0145399, US 18/350,148, CN 202311408257.6

### **Neural network processing method**

Won Woo Ro, Hongju Kal, Cheonjun Park, Hyunwuk Lee, Ipoom Jeong, and **Jiwon Lee**

KR 10-2022-0041848

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**Controller, computing system including the same, and method of creating and searching page table entry for the same**

Seongil O, Won Woo Ro, William Jinho Song, and **Jiwon Lee**

KR 10-2021-0054195, US 17/526,391, CN 202210082121.X

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## ON-GOING RESEARCH – PROJECT NAMES ONLY

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**Enhancing Unified Virtual Memory for AI Workloads (Under Review)**

**Efficient Memory Management of Large-Scale Memory Space (Under Review)**

Accelerating Sparse Matrix Computations

Exploring Use Cases of CXL Memory Devices

Application of IOMMU Specifications

Identifying Bottlenecks in Serverless Environments

Analysis on Different Sparse Matrix Formats

Improving Unified Virtual Memory in GPUs

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## PROJECTS

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**Development of High Performance Multi-GPU Memory System**

Mar. 2021 - Feb. 2024

Research project at Yonsei University joint with National Research Foundation of Korea

**Development and Application of DRAM-based Memory Hierarchies**

Sept. 2020 - Aug. 2023

Research project at Yonsei University joint with Samsung Electronics

**Programmable Neural Network Processors**

June 2018 - May 2022

Research project at Yonsei University joint with Samsung Research Funding and Incubation Center for Future Technology

**PIM Core Technology and System Development for Data-intensive Applications**

Mar. 2018 - Mar. 2020

Research project at Yonsei University joint with Samsung Electronics

**Development of Reconfigurable Artificial Neural Network Accelerator and Instruction Set Architecture**

Mar. 2018 - Feb. 2019

Research project at Yonsei University joint with Korea Electronics Technology Institute

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## ACADEMIC ACTIVITIES

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**External Review Committee**

ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS), Summer Cycle 2026

**Reviewer**

ACM Transactions on Architecture and Code Optimization (TACO), 2025

**Mentorship**

Embedded Systems and Computer Architecture Lab, Yonsei University

Industry-Academic Mentoring Program, Nov. 2024 - Present

Intelligence System and Parallel Computer Architecture Lab Ewha Womans University

Industry-Academic Mentoring Program, Jan. 2025 - Present

**Teaching Assistant**

EEE6611, System Design and Applications Lab, Yonsei University, Fall 2019

EEE4473, Embedded System Lab, Yonsei University, Spring 2019

EEE3535, Operating Systems, Yonsei University, Fall 2018

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## SCHOLARSHIPS AND AWARDS

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**Memory Division Graduate Fellowship**

Samsung Electronics

Sept. 2023 - Aug. 2024

**Excellent Graduate Researcher Scholarship**

Yonsei University

Feb. 2023

**National Scholarship for Science and Engineering**

Korea Student Aid Foundation (KOSAF)

Mar. 2014 - Feb. 2018

**Academic Excellence Award**

Yonsei University

Spring 2014